JIYUN BAE

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RESEARCH INTEREST

My research focuses on **information selection and utilization for intelligent agents**. I aim to understand how AI systems can identify relevant information, retain it efficiently, and leverage it effectively for reasoning and decision-making. My broader interests include robust reasoning, agent memory, personalization, and responsible information attribution.

EDUCATION

Ph.D. in Artificial Intelligence Mar. 2026 – Current
Pohang University of Science and Technology (POSTECH)
Advisor: Jaeho Lee

M.S. in Artificial Intelligence Feb. 2024 – Feb. 2026
Pohang University of Science and Technology (POSTECH)
Advisor: Jaeho Lee

B.E. in Information Security Mar. 2020 – Feb. 2024
Seoul Women's University

PUBLICATIONS

Understanding the Effects of Distractors on Reasoning Vision-Language Models 2026
Jiyun Bae, Hyunjong Ok, Sangwoo Mo, Jaeho Lee
*Empirical Methods in Natural Language Processing (**EMNLP**), 2026, Oral Presentation*

- Introduced *Idis*, a 277K-example VQA benchmark that systematically varies distractor modality, number, and semantics across perception and reasoning-centric tasks.
- Identified a modality-dependent failure mode in test-time scaling: visual distractors degrade accuracy without lengthening reasoning, whereas textual distractors intensify inverse scaling.
- Showed via attribute-level trace analysis and attention-blocking experiments that distractor attributes verbalized in reasoning traces causally drive incorrect answers, motivating a prompt-based mitigation.

WORK EXPERIENCE

Ericsson-LG | *Software Developer Intern* Jan. 2022 – Feb. 2022

- Wrote C++ unit tests (Google Test) for the SMF node's HTTP client component in a commercial 5G Core network codebase, contributing code through Gerrit-based review in a Scrum R&D team.

S2W | *NLP Researcher Intern* Jul. 2022 – Sep. 2022

- Built a category-hierarchy crawler (BeautifulSoup) that collected ∼7.9K cybersecurity Wikipedia articles, empirically capping traversal depth to prevent topic drift into irrelevant pages.

PROFICIENCIES

Languages: Korean (Native), English (Professional)
Machine Learning Libraries: PyTorch (Advanced), TensorFlow (Advanced)

ACTIVITIES

Teaching Assistant | CSED554 Deep Learning for Natural Language Processing, Spring 2025
Lead and Founder | Google Developer Groups on Campus (GDGoC) @ Seoul Women's University, Aug 2022 – Jul 2023
Invited Talk | UMich NLP reading group, Presented EMNNLP2026 paper, Oct 2026

REFERENCES

Jaeho Lee, Associate Professor at POSTECH | Ph.D. Advisor
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Sangwoo Mo, Assistant Professor at POSTECH | Research Collaborator
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